Proximal Gradient

$\min \quad F(x) = f(x) + g(x)$ $\quad f$ Lip-Smooth. Nonconvex

Non-smooth g LSC/ Closed

$$x^{k+1} = \operatorname{argmin} \; F(x) + \underbrace{\frac{L}{2}\|x - x^k\|^2}_{\text{Proximal term}} = \operatorname{Prox}_{\frac{1}{L}F}(x^k)$$

Linearized f

$$x^{k+1} = \operatorname{argmin} \; f^k + \langle x - x^k, \nabla f^k \rangle + g(x) + \frac{c}{2}\|x - x^k\|^2$$

$$= \operatorname{argmin} \; \frac{1}{c} g(x) + \frac{1}{2}\|x - (x^k - \frac{1}{c}\nabla f^k)\|^2$$

$$= \operatorname{Prox}_{\frac{1}{c}g}(x^k - \frac{1}{c}\nabla f^k) \qquad c = \gamma L. \quad \gamma > 1$$

当 $g(x) = I_C(x)$ 即 Gradient Projection

$$x^{k+1} = \operatorname{Proj}_C(x^k - \alpha_k \nabla f^k)$$

当 $g(x) = \lambda\|x\|_1$ 即 ISTA $\quad \min f(x) + \lambda\|x\|_1$

$$x^{k+1} = \text{Soft_Thrd}_{\lambda/L}(x^k - \frac{1}{L}\nabla f^k)$$

e.g. ISTA for LASSO $\quad x^{k+1} = \text{Soft_Thrd}_{\lambda/\|A\|^2}\{x^k - \alpha A^H(Ax - y)\}$

Convergence

F 非凸，可收敛于 Local min

f L-smooth

$\downarrow$

Descent Lemma

$x^{k+1} = \operatorname{argmin}_x \sim$

代入 x 为 x^k

$\downarrow$

Sufficient Descent Property

$$F^{k+1} + \frac{1}{2}\|\Delta x^k\|^2 \le F^k$$

$\downarrow$ $\sum_{k=1}^{K}\|\Delta x^k\| \le F^k - F^0 < +\infty$

Interval $\Delta x^k \to 0$

F coercive

$\downarrow$

x^k Bounded

$\downarrow$

Subseq $x^{k_i} \to \tilde{x}$

$\downarrow$ Wasserstrain

$x^{k+1} = \underset{x}{\operatorname{argmin}} \sim$ $\xrightarrow{\limsup \geq}$ $F(x^{k_i}) \to F(\tilde{x})$ $\xleftarrow{\liminf \geq}$ g LSC

代入 x 为 $\tilde{x}$ 代入 x^{k+1} 为 x^{k_i}

$x^{k+1} = \underset{x}{\operatorname{argmin}} \sim$

Opt Cond. $0 \in \partial \sim$

$\tilde{x} \in \operatorname{Crit} F \leftarrow$ Gradient

$\|\partial F^k\| \to 0$

KL Property $\downarrow$

Seq $x^k \to \tilde{x}$

1. Sufficient Descent Property $F^{k+1} \leq F^k$

即 Value Descent 值收敛 $F \to \tilde{F}$

假设 f Lip-Cnt，由 Descent Lemma 得

$$f^{k+1} \leq f^k + \langle \nabla f^k, x^{k+1} - x^k \rangle + \frac{L}{2} \|x^{k+1} - x^k\|^2 \quad ①$$

$$x^{k+1} = \operatorname{argmin} f^k + \langle x - x^k, \nabla f^k \rangle + g(x) + \frac{c}{2} \|x - x^k\|^2$$

代入 x 为 x^k

$$\langle x^{k+1} - x^k, \nabla f^k \rangle + g^{k+1} + \frac{c}{2} \|x^{k+1} - x^k\|^2 \leq g^k \quad ②$$

① + ② 得

$$F^{k+1} \leq F^k + \frac{1}{2}(L - c) \|x^{k+1} - x^k\|^2$$

$$= F^k + \frac{1}{2}(1 - \gamma) L \|x^{k+1} - x^k\|^2 \leq F^k$$

2. Gradient Convergence 梯度收敛 $\partial F^k \to 0$

$$x^{k+1} = \operatorname{argmin} \frac{L}{2} \|x - x^k\|^2 + \langle x - x^k, \nabla f^k \rangle + g(x)$$

最优条件 $0 \in L(x^{k+1} - x^k) + \nabla f^k + \partial g^{k+1}$

$$\partial F^{k+1} = \nabla f^{k+1} + \partial g^{k+1}$$

$$= -L(x^{k+1} - x^k) + \nabla f^{k+1} - \nabla f^k$$

$$\|\partial F^{k+1}\| \leq \|L(x^{k+1} - x^k)\| + \|\nabla f^{k+1} - \nabla f^k\|$$

$$\leq 2L \|x^{k+1} - x^k\| \quad \leftarrow \text{Lipschitz}$$

故 $\partial F(x^k) \to 0$

3. Interval Convergence $\|x^{k+1}-x^k\| \to 0$

将 $F(x^k)-F(x^{k+1}) \geq \frac{1}{2}(c-L)\|x^{k+1}-x^k\|^2$

前 k 步求和为有限数

$$\frac{1}{2}(c-L)\sum_{k=1}^{K}\|\Delta x^k\|^2 \leq F(x^0)-F(x^K) \leq F(x^0)-\tilde{F} < \infty$$

故 $\|\Delta x^k\|^2 \to 0$

4. Subsequence Convergence 子列收敛 $x^{k_i} \to \tilde{x}$

$\|x^{k+1}-x^k\|^2 \to 0$

由于 F 不一定 coercive（如 $g(x)=\|x\|_0$）

$\{x^k\}$ 有界 bounded 需额外假设

Wasserstrass Thm. 若 $\{x_k\}$ 有界，则

必有收敛子列 Subsequence $\{x^{k_i}\} \to \tilde{x}$ 极限点

5. Subsequence Value Convergence 子列值收敛 $F(x^{k_i}) \to F(\tilde{x})$

序列收敛不能说明值收敛

光滑部分可以 $\lim_{i\to\infty} f(x^{k_i}) = f(\lim_{i\to\infty} x^{k_i}) = f(\tilde{x})$

↳ f 连续

非光滑部分

$$x^{k+1} = \arg\min \frac{L}{2}\|x-x^k\|^2 + \langle x-x^k, \nabla f^k\rangle + g(x)$$

故 $\frac{L}{2}\|x^{k+1}-x^k\|^2 + \langle x^{k+1}-x^k, \nabla f^k\rangle + g(x^{k+1})$

$$\leq \frac{L}{2}\|\tilde{x}-x^k\|^2 + \langle \tilde{x}-x^k, \nabla f^k\rangle + g(\tilde{x})$$

令 k+1 为子列标号 k_i，再让两边 $i\to\infty$

$$\underbrace{0+0+}_{\text{因为 } \|x^{k+1}-x^k\|\to 0} \lim_{i\to\infty}\sup g(x^{k_i}) \leq \underbrace{0+0+}_{\text{因为子列 } x^{k_i}\to\tilde{x}} g(\tilde{x})$$

同时，由于 g 为 LSC，$\lim_{i\to\infty}\inf g(x^{k_i}) \geq g(\tilde{x})$

故 $g(x^{k_i}) \to g(\tilde{x})$ 即 $\lim_{i\to\infty} F(x^{k_i}) = F(\tilde{x})$

6. Limit Point 即 Critical Point $\tilde{x} = x^*$

根据 Frechet Subdiff 的定义

$$\hat{\partial} F(x) := \{ g \mid F(y) \geq F(x) + \langle g, y-x \rangle + o(y-x), \forall y \}$$

或 $g \in \hat{\partial} F(x)$ 满足 $\lim\limits_{y \neq x} \inf\limits_{y \to x} \dfrac{F(y) - F(x) - \langle g, y-x \rangle}{\| y - x \|} \geq 0$

Limiting Subdiff 的定义

假设 $x^k \to \tilde{x}$ 且 $F(x^k) \to F(\tilde{x})$

若 $g^k \in \hat{\partial} F(x^k) \to g$，则 $g \in \partial F(x^k)$ 为 Limiting Subdiff

由 $\partial F(x^{k_i}) \to 0$ 和 $x^{k_i} \to \tilde{x}$，根据定义 $\partial F(\tilde{x}) = 0$

（第2条）　　（第5条）

即 $\tilde{x} = x^*$ 或记作 $\tilde{x} \in \operatorname{crit} F$

7. Sequence Convergence 序列收敛 $x^k \to \tilde{x}$

若 F 满足 KL Property，则 x^k 为 Cauchy 列 $x^k \to x^*$